9^{th} World Mathematics Team Championship 2018

Junior Level Individual Round 1

English Version

Instruction: This round has 15 questions (**20 minutes**).
Each question is worth 2 points.
No point penalty for submitting wrong answer.
Blank answer will be assigned 0.5 point.

1. Find $1 + 2 – 3 + 4 + 5 – 6 + 7 +8 – 9 + \ldots + 97 + 98 – 99$.

A) 1684 B) 1724 C) 1584 D) 1592 E) 1846

2. If $1-\dfrac{1}{1+\dfrac{1}{1-\dfrac{1}{1+\dfrac{1}{2}}}}=\dfrac{p}{q}$ where p and q are relatively prime find $p + q$.

A) 7 B) 6 C) 5 D) 4 E) 3

3. Compute
$(11 + 10 + ... + 1) + (10 + 9 + ... + 2) + (9 + 8 + ... + 3) + (8 + ... + 4) + (7 + 6 + 5) + 6$.

A) 324 B) 284 C) 216 D) 188 E) 144

4. Find $\dfrac{2^3+2^4+2^5}{3^2+4^2+5^2}$.

A) $\dfrac{28}{25}$ B) $\dfrac{14}{12}$ C) $\dfrac{11}{10}$ D) $\dfrac{6}{5}$ E) $\dfrac{27}{25}$

5. If $A(x,y)=\dfrac{x}{y}+\dfrac{y}{x}$, find $A(1,2)+A(1,3)+A(1,4)+A(2,3)+A(2,4)+A(3,4)$.

A) 12 B) $13\dfrac{1}{6}$ C) $13\dfrac{5}{6}$ D) $16\dfrac{5}{6}$ E) 15

6. The least common multiple of 300 and 2520 equals:

A) $2^4\times3^3\times5^2\times7$ B) $2^2\times3^3\times5^2\times7$ C) $2^2\times3^2\times5^2\times7^2$

D) $2^3\times3^2\times5^3\times7$ E) $2^3\times3^2\times5^2\times7$

7. Find $\left(1-\dfrac{1}{2}\right)\left(1-\dfrac{1}{3}\right)\left(1-\dfrac{1}{4}\right)\ldots\left(1-\dfrac{1}{2018}\right)$.

A) $\dfrac{1}{2017}$ B) $\dfrac{1}{2018}$ C) $\dfrac{2}{2017}$ D) $\dfrac{2018}{2017}$ E) $\dfrac{2017}{2018}$

9^{th} World Mathematics Team Championship 2018

8. Find the product of the smallest and the largest of the fractions $\frac{9}{8}, \frac{4}{7}, \frac{13}{12}, \frac{5}{9}, \frac{2}{3}$.

A) $\frac{13}{21}$ B) $\frac{13}{18}$ C) $\frac{3}{4}$ D) $\frac{5}{8}$ E) $\frac{9}{14}$

9. Find the value of $\frac{2\times 33\times 404\times 5005}{6\times 77\times 808\times 9009}$.

A) $\frac{10}{63}$ B) $\frac{5}{126}$ C) $\frac{5}{84}$ D) $\frac{15}{196}$ E) $\frac{3}{216}$

10. Find the last digit of $1\times 2+2\times 3+3\times 4+...+2017\times 2018$.

A) 2 B) 4 C) 6 D) 8 E) 9

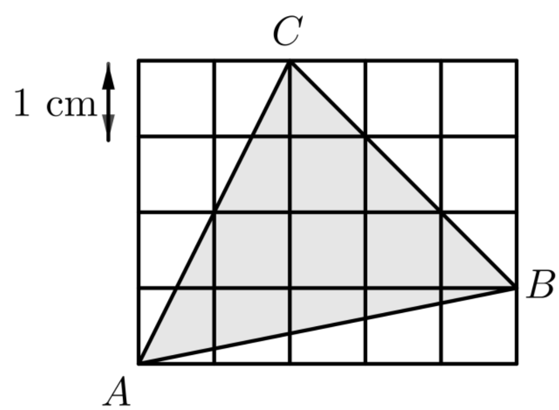
11. The greatest common divisor of 1452 and 2970 equals:

A) 66 B) 242 C) 36 D) 6 E) 44

12. Find 2018-th digit after decimal point of the fraction $\frac{123}{7}$.

A) 5 B) 7 C) 1 D) 4 E) 2

13. Find the area of triangle *ABC*.



A) 10 cm^2 B) 9.5 cm^2 C) 9 cm^2 D) 8.5 cm^2 E) 8 cm^2

14. For any two integers *a* and *b*, let $a \heartsuit b = a \times b + 3a$. Find $(2\heartsuit 0)\heartsuit(1\heartsuit 8)$.

A) 112 B) 84 C) 200 D) 162 E) 76

15. If *a*, *b*, *c* and *d* are 3, 4, 5, 6 in some order find the least value of $\frac{a}{b}+\frac{c}{d}$.

A) $\frac{1}{2}$ B) $\frac{4}{3}$ C) $\frac{19}{15}$ D) $\frac{6}{5}$ E) $\frac{9}{5}$

9th World Mathematics Team Championship 2018

Junior Level Individual Round 2

English Version

Instruction: This round has 8 questions (**40 minutes**).

Question numbers 1, 2, 3, 4, 5 and 6 are worth 4 points each.

Question numbers 7 and 8 are worth 8 points each.

No point penalty for submitting wrong answers.

1. Car with 4 wheels and 1 spare tire travelled for 1000 kilometers. Each of two of the tires has been used for 500 kilometers. The other three tires have been used equal number of kilometers. Find this number.

2. For how many positive integer values of a the 12 numbers

$$a,\ 1,\ 2^1,\ 2^2,\ \ldots,\ 2^{10}$$

can be split into two sets of equal sums?

3. Six integers x, $x + 1$, $x + 2$, $x + 3$, $x + 4$ and $x + 5$ are such that the sum of any two of them equals to one of the remaining numbers or can be expressed as a sum of two or three from the remaining numbers. How many such x exist?

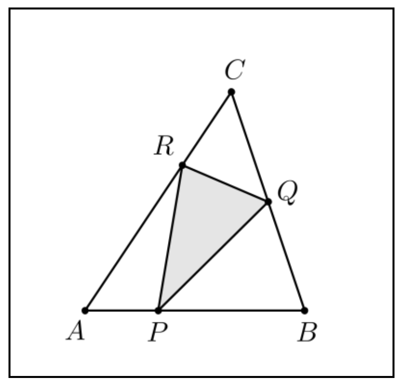
4. Find the number of three digit numbers such that the difference between any two neighboring digits is not 1.

9^{th} World Mathematics Team Championship 2018

5. We have 8 colors. Find the number of ways one can color the cells of 2×2 table such that no two adjacent cells are of the same color.

(A color may be used more than once. Two cells are adjacent if they share a common side.)

6. Points P, Q and R on the sides AB, BC and CA of triangle ABC are such that $AP:PB = 1:2$; $BQ:QC = 1:1$ and $CR:RA = 1:2$. If $S_{ABC} = 18\text{ cm}^2$, find S_{PQR}.



---

7. Find the number of solutions of $\dfrac{a}{b}+\dfrac{c}{d}=\dfrac{e}{f}$ where a, b, c, d, e, and f are 1, 2, 3, 4, 5 and 6 in some order.

8. A person is born in year $\overline{abcd}$. A year $\overline{mnpq}$ is good for that person if after his birthday his age in this year equals $a+b+c+d$. For example, 2017 is good for a person born in 2012. How many years are there between year 2000 and year 2018 inclusive that are not good for anyone?

Junior Level Relay Round 1A

R1-A A three-digit number is called nice if it coincides with any of the numbers 135, 246 and 987 in exactly one position (for example 147 and 285 are nice but 239 and 487 are not). How many three-digit numbers are nice?

Junior Level Relay Round 1B

R1-B T = TNYWR (The Number You Will Receive). Rectangle *ABCD* on the picture is divided into 9 smaller rectangles. The areas of some of the rectangles are given. Find the area of the rectangle *ABCD*.

3	6	3
		2
	T	

Junior Level Relay Round 2A

R2-A In Bulgaria we have coins of 1, 2, 5, 10, 20 and 50 stotinki. What is the minimum number of coins in a set *A* such that any amount from 1 to 100 stotinki can be expressed as several coins from *A*?

Junior Level Relay Round 2B

R2-B T = TNYWR (The Number You Will Receive). Points *M* and *N* are midpoints of the sides *BC* and *CD* of quadrilateral *ABCD*. If $S_{ABD} = T$ cm^2 and $S_{BCD} = 2T$ cm^2, find S_{AMN}.

Junior Level Relay Round 3A

R3-A Eight teams played in a football tournament. Every two teams played one game against each other (win 3 points, draw 1 point, lost 0 points). Total number of points gained by the 8 teams equals 79. How many games ended by draws?

Junior Level Relay Round 3B

R3-B T = TNYWR (The Number You Will Receive). The number $\overline{a2018b}$ is divisible by 11 and the number $\overline{aT}$ is divisible by 3. Find the maximum value of $a + b$.

9^{th} World Mathematics Team Championship 2018

Junior Level Team Round

English Version

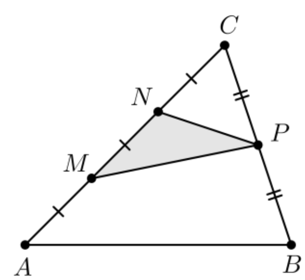
Instruction: This round has 14 questions (**40 minutes**).
Each question is worth 5 points.
No point penalty for submitting wrong answer.

1. Find the number of 3 digit numbers without 0 in their decimal representation with sum of all digits equals to 23.

2. In a pile of 9 coins one is a fake. Suppose we can find out whether the fake coin is among them in any move where we pick four coins. Find the minimum number of moves that guarantee finding the fake coin.

3. The average weight of a group of people is 35.2 kilograms. Albert, who weighs 45.6 kilograms, then joins the group. This raises the average weight of the group to 36 kilograms. How many children were in the original group?

4. Points M, N on the side AC and point P on the side BC of triangle ABC are such that $AM = MN = NC$ and $BP = PC$. If $S_{ABC} = 30\text{ cm}^2$, find S_{MNP}.



<u>9th World Mathematics Team Championship 2018</u>

5. Four consecutive positive integers have the following property:

The first number is divisible by 3

The second number is divisible by 5

The third number is divisible by 7

The fourth number is divisible by 9

Find the least possible sum of these numbers.

6. How many numbers in the interval [1, 2018] can be expressed as a sum of k consecutive positive integers for each $k = 2$, $k = 3$ and $k = 5$?

7. Find the number of sequences of zeros and ones of length 4.

8. Each cell of a 4×4 table is colored either white or black. Any white cell has exactly 3 black neighboring cells and any black cell has exactly one white neighboring cell. Find the number of white cells. (Two cells are neighbors if they share a common side.)

9. If the number $\overline{a2018b}$ is divisible by 12 but not divisible by 9 find the largest value of $a + b$.

9th World Mathematics Team Championship 2018

T = The Solution of Problem #2

10. Two candles of equal height burn completely for 3 hours and for 2 hour, respectively. The two candles have been lighted simultaneously. After how many minutes the height of one of them is $\boldsymbol{T}$ times the height of the other?

T = The solution of problem #9

11. The numbers 4, 5, 6, 7, 8, 9, 10, 11 and 12 are written on 9 balls. Every ball is either blue or red. The average of the numbers on blue balls is 7, the average of the numbers on red balls is $\boldsymbol{T} - 3$. Find the number of blue balls.

T = The Solution of Problem #8

12. A point M on the side AB of a parallelogram $ABCD$ is such that $AM = \boldsymbol{T} \cdot MB$. If $S_{CDM} = 25\ \text{cm}^2$, find S_{AMD}.

T = The Solution of Problem #5

13. All positive integers are written one after another 123456789101112… . Find the digit on the $\boldsymbol{T}$-th place in this sequence.

S = The Solution of Problem #7; T = The Solution of Problem #4

14. A house consists of several rooms. Each room has $\boldsymbol{T}$ doors. Two of the doors are outside and the remaining doors are between rooms. If the total number of doors is $\boldsymbol{S}$, how many rooms are in the house?